

CE 310 — Week 9 Pre-Lab Quiz

Week: 9 | Topic: Simple Linear Regression — Concepts and Python/scipy Points: 10 questions × 1 pt each = 10 pts Submission: Complete in D2L before class begins. No partial credit for MC.

Instructions

Select the single best answer for each question. The quiz covers simple linear regression, the role of R^2 , residuals, and the Python tools you will use in today's lab. Review the Week 9 Background reading before completing this quiz.

Questions

Q1. Simple linear regression gives you an equation of the form $\hat{y} = \beta_0 + \beta_1 \cdot x$. What does this equation allow you to do that correlation alone does not?

- (A) Measure how strong the linear relationship is between two variables
 - (B) Test whether two variables are statistically independent
 - (C) Predict a specific numerical value of y for any given value of x
 - (D) Determine whether the relationship between x and y is causal
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Q2. In the regression equation $\hat{y} = \beta_0 + \beta_1 \cdot x$, the slope $\beta_1 = 1.16 \text{ kWh/}^\circ\text{F}$ for a building's cooling model. Which statement correctly interprets this slope?

- (A) When outdoor temperature is 0°F , the building uses 1.16 kWh of cooling
 - (B) For every 1°F increase in outdoor temperature, predicted cooling demand increases by 1.16 kWh/hr
 - (C) Cooling demand increases 1.16 times faster than outdoor temperature
 - (D) The building uses 1.16 kWh of cooling on an average day
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Q3. $R^2 = 0.60$ for a regression model predicting cooling energy from outdoor temperature. Which statement is the most accurate engineering interpretation?

- (A) 60% of outdoor temperature variation is explained by cooling energy
 - (B) The regression line passes through 60% of the data points
 - (C) Outdoor temperature explains 60% of the variance in cooling energy; 40% is explained by other factors (time of day, occupancy, equipment loads)
 - (D) The model is 60% accurate and should not be used for design decisions
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Q4. In Weeks 5–7 you worked with the DOE Reference Medium Office dataset for Tucson (8,760 hourly observations). You computed the Pearson correlation between Outdoor_Temp_F and Cooling_kWh using Excel's CORREL function. Approximately what value did you find?

- (A) $r \approx 0.23$ (weak positive)
 - (B) $r \approx 0.60$ (moderate positive)
 - (C) $r \approx 0.78$ (strong positive)
 - (D) $r \approx 0.99$ (nearly perfect)
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Q5. Using the same DOE dataset, which statement about minimum cooling energy is physically correct?

- (A) The minimum cooling value is negative, indicating the chiller removes heat from outside
 - (B) The minimum cooling value is 0 kWh, because the chiller turns off when cooling is not needed
 - (C) The minimum cooling value equals the mean cooling value divided by 24
 - (D) The minimum cooling value occurs at the highest outdoor temperature
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Q6. A residual is defined as:

- (A) The slope of the regression line at a given data point
 - (B) The difference between the observed y -value and the value predicted by the regression model: $e_i = y_i - \hat{y}_i$
 - (C) The square root of R^2
 - (D) The standard deviation of the predictor variable x
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Q7. A colleague says “the regression line and the correlation coefficient tell you the same thing.” Which response is correct?

- (A) True — correlation r and the regression slope β_1 are identical
 - (B) True — $R^2 = r$, so they convey the same information
 - (C) False — correlation measures strength and direction; regression gives you the actual predictive equation including slope, intercept, and units
 - (D) False — correlation applies only to Normal data; regression applies only to skewed data
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Q8. Linear regression is the foundation of which machine learning category?

- (A) Unsupervised learning (finding hidden patterns in unlabeled data)
- (B) Supervised learning (learning a mapping from inputs to labeled outputs)
- (C) Clustering (grouping similar observations)

(D) Dimensionality reduction (compressing many features into fewer)

Q9. In Python, `scipy.stats.linregress(x, y)` returns five values. Which of the following is the correct complete list?

- (A) slope, intercept, r_value, p_value, std_err
 - (B) slope, intercept, r_squared, p_value, confidence_interval
 - (C) beta_0, beta_1, pearson_r, F_statistic, residual_std
 - (D) slope, intercept, r_value, t_statistic, degrees_of_freedom
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Q10. In pandas, `df['A'].corr(df['B'])` computes:

- (A) The covariance between columns A and B, in units of $A \times B$
 - (B) The Pearson correlation coefficient r between columns A and B, always between -1 and $+1$
 - (C) The slope of the regression line with A as predictor and B as outcome
 - (D) The R^2 value for a regression of B on A
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